

LICENSE

InfantCryNet — Multimodal Clinical Inference Engine

Copyright © 2026 Safia Benarbia

Djillali LIABES University | safiabenarbia.2001@gmail.com

Software License — MIT

MIT License — Full Text

MIT License

Copyright (c) 2026 Safia Benarbia

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the “Software”), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED “AS IS”, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Medical Disclaimer & Limitation of Liability

Medical Disclaimer & Limitation of Liability

1. Academic Research Purpose Only

InfantCryNet is an **academic research project** developed in partial fulfilment of the requirements for a Master in Applied Mathematics (Probability and Statistics) at Djillali LIABES University. It is **not** a certified, approved, or validated medical device under any applicable regulatory framework, including but not limited to:

- the European Union Medical Device Regulation (**EU MDR 2017/745**);
- the U.S. Food and Drug Administration digital health guidance (**FDA 510(k) / De Novo**);
- the Algerian National Agency for Pharmaceutical Products and Medical Devices (**ANPP**).

This software has **not** undergone clinical validation trials, regulatory review, or safety certification of any kind.

2. No Clinical Diagnosis — Absolute Prohibition

This software must not be used, directly or indirectly, for:

- (a) making or supporting **clinical diagnoses** of any medical condition in any infant, child, or human being;
- (b) **life-or-death decisions** concerning the medical care, treatment, or welfare of an infant;
- (c) **replacing, supplementing, or informing** the professional clinical judgment of a licensed medical doctor, paediatrician, nurse, or any qualified healthcare provider;
- (d) **deployment in any clinical, hospital, neonatal, or patient-care environment** without explicit regulatory clearance from the competent national authority.

Any use of this software in a clinical context, with or without modification, is **entirely at the risk of the user** and constitutes a violation of the terms of this licence.

3. No Warranty of Medical Accuracy

The probability estimates, confidence scores, conformal prediction sets, clinical tier labels, and all other outputs produced by this system are **research artefacts**, not clinical assessments. They are derived from models trained on a limited single-site dataset under a specific IRB protocol (CHU-SBA-2025-017) and **may be incorrect, biased, or inapplicable** to any individual infant or clinical population.

The author, **Safia Benarbia**, expressly **disclaims all warranties**, whether express, implied, statutory, or otherwise, including any warranty that the software or its outputs are medically accurate, fit for clinical use, or suitable for any purpose involving human health.

4. Limitation of Liability — Maximum Extent Permitted by Law

To the fullest extent permitted by applicable law, in no event shall **Safia Benarbia**, her university, supervisors, collaborators, or affiliated institutions be liable for:

- any direct, indirect, incidental, special, consequential, or exemplary damages;
- any harm, injury, death, distress, or adverse outcome to any infant or human being arising from reliance on the outputs of this software;
- any loss of data, loss of revenue, or reputational damage;
- any regulatory or legal liability incurred by a user who deploys this software outside the expressly permitted research context.

This limitation applies regardless of the theory of liability (contract, tort, negligence, strict liability, or otherwise) and even if the author has been advised of the possibility of such damages.

Simulated Data Declaration

Simulated Data Declaration

The following data and signals used within this repository are **entirely synthetic and computationally simulated**. They do not represent, reproduce, or approximate any real patient, infant, or clinical recording beyond the statistical distributions described in the accompanying research paper:

Data Type	Status	Source
Video features (50-D)	Simulated	<code>video_simulator.py</code>
Bio context inputs	Simulated	<code>bio_model.py</code>
Demo audio probs	Simulated	<code>run_demo.py</code>
Benchmark scenarios	Simulated	<code>synthetic_benchmark.py</code>
Raw clinical audio	Restricted	CHU-SBA vault (Restricted)
Raw clinical video	Restricted	CHU-SBA vault (Restricted)

The simulated video features are generated using a stochastic prototype model grounded in published infant motor-pattern norms (Gustafson & Harris, 1990; Zeifman, 2001). These simulations **do not constitute clinical data** and should not be interpreted as measurements of any real infant.

The biological context signals (feeding intervals, temperature proxies, sleep duration estimates) are generated by a rule-based expert system derived from publicly available paediatric guidelines (WHO/UNICEF, AAP, NICE CG149). They are **not** real patient measurements.

The real clinical data (audio recordings and annotations) used to train and evaluate the acoustic branch was collected under ethics protocol and remains in the exclusive custody of CHU Sidi Bel Abbès. It is **not distributed** in this repository and is not accessible to the public.

User Responsibility Statement



User Responsibility Statement

By downloading, cloning, installing, executing, or otherwise using any part of this software, the user **explicitly accepts** the following:

1. **Full personal responsibility** for any and all consequences, outcomes, decisions, or actions that result from their use of this software, including any use that violates the terms of this licence.
2. **Acknowledgement that this is research software**, not a medical device, and that it must not be used in any clinical, diagnostic, therapeutic, or patient-care context without appropriate regulatory clearance.
3. **Compliance with applicable law**: the user agrees to comply with all local, national, and international laws and regulations governing the use of AI systems in healthcare, including medical privacy laws, data protection regulations, and medical device regulations in their jurisdiction.
4. **IRB compliance**: any researcher applying this software to human-subjects data (including infant audio or video recordings) must obtain independent Institutional Review Board (IRB) or Ethics Committee approval from their own institution before doing so.
5. **No indemnification claim**: the user waives any right to seek indemnification, contribution, or compensation from Safia Benarbia, her university, or any affiliated institution for any harm arising from their use of this software.

If you are a clinician, nurse, parent, or caregiver seeking advice about a crying infant, please consult a qualified medical professional. This software cannot replace that advice.

Contact & Attribution

Author	Safia Benarbia
Institution	Djillali LIABES University, Master in Probability and Statistics, Algeria.
Contact	safiabenarbia.2001@gmail.com
Repository	github.com/safiabenarbia2001-ux/InfantCryNet
Ethics Protocol	CHU-SBA-2025-017
Year	2026
License	MIT (code only) — see Section 1 above
Data	All clinical data remains at CHU-SBA (not distributed)

This document constitutes the complete licence and disclaimer for InfantCryNet v1.0. It supersedes any informal statements made in code comments, README files, or oral presentations.

“The algorithm is open. The data is protected. The responsibility is shared fairly.”

Safia Benarbia — 2026